

Q1. Write a Python program to create an age_records DataFrame containing the attributes Person_ID and Age for suitable sample records. Perform Data Discretization on the Age attribute using Equal Frequency Binning. Display the original and discretized data.

[10 Marks]

Q2. Create a DataFrame containing information about different cities with the attributes City, Area (sq. km), Average Temperature (°C), Rainfall (mm), and Population Density. Generate a Treemap to visualize the population density of each city and create a 3D Scatter Plot using Area, Average Temperature, and Rainfall. Add suitable labels, colors, and titles to both visualizations.

[20 Marks]

OR

Q2. Write a Python program to perform market basket analysis on the Groceries dataset using the Apriori algorithm. Read and explore the dataset, preprocess the data by handling missing values, convert categorical transaction data into a suitable numeric format, and generate frequent itemsets and association rules. Repeat the analysis using minimum support values 0.3 and 0.4.

[20 Marks]

S.No.	Items Purchased
1	Basmati Rice, Toor Dal, Turmeric Powder, Cooking Oil
2	Atta, Sugar, Tea, Milk
3	Rice, Moong Dal, Salt, Cooking Oil
4	Wheat Flour, Chickpeas, Turmeric Powder, Cumin Seeds
5	Poha, Peanuts, Onions, Cooking Oil
6	Basmati Rice, Paneer, Garam Masala, Tomatoes
7	Atta, Potatoes, Onions, Cooking Oil

Q3. Viva

[5 Marks]